Beginner to ICPC Regionalist

How I started CP and what resources I followed

- **ICPC Regionalist** 2023-2024 (2 Regionals qualified)
- **Expert** at Codeforces
- Codechef - **4 star**
- Leetcode - **Knight**

Standard Library (STL) & Basic Programming

Channels & Resources you can follow:-

- LUV Playlist
- CSES Problem set

Topic:-

- Hashmap
- Heap
- Set
- Pairs
- Complexity Analysis
- Stack, Queue,
- Sorting

Number Theory

Channels & Resources you can follow:-

- Utkarsh Gupta Number Theory
- CP Algorithm Algebra Section

1) Sieve of Eratosthenes
2) Fibonacci Numbers
3) Binary Exponentiation
4) Modular Inverse
5) Chinese Remainder theorem
6) LCM, GCD and maths knowledge

Binary Search

Channels & Resources you can follow:-

- Priyansh Agarwal Binary Search Video

- CP Algorithms

- Basic Binary Search Implementation
- Some inbuilt functions:- Lower Bound and Upper Bound
- Ternary Search

Greedy Algorithm + Some Important Topics

Channels & Resources you can follow:-

- Algorithms Live Greedy Problem videos (with proofs)
- GFG Problem Section on Greedy Algorithm

Some Other Important Topics:-

- Prefix & Suffix Sum
- Kadane’s Algorithm
- Two Pointers
- Sliding Window

Dynamic Programming

Channels & Resources you can follow:-

- Aditya Verma DP Playlist
- Errichto's Beginner Videos for DP
- CSES Problem Sets

- Top Down (Recursion + Memoization) & Bottom - Up (Tabulation)
- DP on Arrays
- DP on Grids
- DP on Strings
- DP on Trees.

Graph Theory

Channels & Resources you can follow:-

- Striver Graph Videos
- Galen Collins Graph Videos

- DFS
- BFS
- Topological Sort
- Disjoint Subset Union (DSU)
- Dijkstra
- Floyd Warshall